
Technical Specification

BizBook — Online Booking System for Barbershops

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1. Project Overview

BizBook is a web-based online booking system designed for barbershops.

The system allows clients to book services online and enables barbershop owners to manage appointments, services, staff, schedules, and basic business data through an admin panel.

The main goal of the project is to replace manual booking methods (phone calls, messages, notebooks) with a centralized digital solution.

2. System Architecture

BizBook follows a **client-server architecture** using a modern frontend framework and a Backend-as-a-Service (BaaS) platform.

High-level architecture:

- **Frontend:** Web application (client & admin interfaces)
- **Backend:** Supabase (BaaS)
- **Database:** PostgreSQL (managed by Supabase)
- **Authentication:** Supabase Auth
- **Hosting / Deployment:** Figma Site (demo), local development via Vite

3. Technology Stack

Frontend

- **Framework:** React
- **Build Tool:** Vite
- **Language:** TypeScript / JavaScript
- **Styling:** Tailwind CSS
- **UI Components:** Custom components generated and adapted from Figma
- **Icons:** Lucide Icons

Backend

- **Platform:** Supabase
- **Role:** Backend-as-a-Service (BaaS)
- **Features used:**
 - Database
 - Authentication
 - API access

Database

- **Type:** PostgreSQL
- **Managed by:** Supabase
- **Main entities:**
 - Services
 - Barbers (Masters)
 - Appointments
 - Clients
 - Products (Marketplace)

4. Core Functional Modules

4.1 Client Booking Module

Allows clients to:

- Choose a booking method (by service or by barber)
- Select a service with price and duration
- Select a barber based on profile and availability
- Choose an available date and time
- Submit contact information (name, phone number)

This module focuses on simplicity and minimal input to reduce friction during booking.

4.2 Admin Panel Module

The admin panel provides centralized management of the barbershop.

Main features:

- Manage bookings (view, confirm, cancel)
- Manage services (add, edit, delete)
- Manage barbers (profiles, activation status)
- Configure working schedules
- Block specific dates or periods
- View basic statistics

The admin interface is designed for non-technical users.

4.3 Scheduling & Availability Module

- Displays available time slots
 - Prevents double bookings
 - Supports working hours and breaks
 - Supports blocked dates (holidays, maintenance)
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4.4 Marketplace Module

The system includes a **basic marketplace interface** where clients can view barbershop-related products.

Current functionality:

- Product catalog display
- Categories and product cards
- Integration with booking flow (visual only)

Note:

Payments and order processing are not implemented and are considered out of scope for the current version.

4.5 Statistics & Analytics Module

Provides basic business insights:

- Total number of bookings
- Booking statuses (confirmed, canceled, pending)
- Simple performance overview

The analytics module is intended as a foundation for future expansion.

5. Authentication & Security

- Admin access is protected via **Supabase Authentication**
- Client booking does not require registration
- Sensitive operations are restricted to authenticated admin users

6. Data Flow Overview

1. Client submits a booking request via the frontend
2. Request is sent to Supabase API
3. Data is stored in the PostgreSQL database
4. Admin panel retrieves and displays updated data in real time

7. Limitations & Assumptions

- The project is a **demo / prototype version**
- No online payments are implemented
- Marketplace is informational only
- Some data may be mock or sample data
- Free-tier Supabase limitations may apply

8. Future Improvements

Possible future enhancements:

- Online payment integration
- SMS / email notifications
- Advanced analytics dashboards
- Multi-branch barbershop support
- Client accounts and booking history

9. Conclusion

BizBook demonstrates how modern web technologies can be used to digitalize traditional service-based businesses.

The system provides a scalable foundation for real-world deployment while remaining simple and user-friendly.